

Extended Abstract

Motivation Reinforcement Learning from Human Feedback (RLHF) fundamentally constrains exploration by relying on static reference policies for response comparisons. While Exploratory Preference Optimization (XPO) theoretically enables flexible sampling, implementations default to original reference sampling, creating a "comparison bottleneck" that limits behavioral diversity and novel capability discovery. This work addresses how adaptive sampling from historical policies could overcome this limitation and synergize with XPO's optimism mechanism.

Method We introduce historical policy sampling for XPO, enabling dynamic selection of comparison responses from past policy snapshots. The approach leverages Low-Rank Adaptation (LoRA) to maintain policy deltas in a fixed-size reservoir, enabling efficient reconstruction of historical behaviors with constant memory overhead. Three strategies are developed:

- *GreedyWorst*: Comparisons against underperforming policies
- *UniformPast*: Broad sampling across historical behaviors
- *DiversityBoost*: Coverage optimization for rare response patterns

This framework transforms training trajectories into explorable behavioral repertoires that complement XPO's optimism.

Implementation & Results Using Qwen 2.5 0.5B on UltraFeedback, our analysis reveals that standard DPO shows minimal improvement over baselines despite significant optimization. Historical sampling creates higher-value-difference comparisons that could yield stronger learning signals, with each strategy offering distinct exploration profiles. Qualitative analysis shows DPO produces safer responses but fails to develop novel capabilities, highlighting opportunities where historical sampling could enhance exploration.

Discussion & Conclusion Historical sampling synergizes with XPO's optimism by:

- Providing diverse behavioral anchors for comparison
- Amplifying exploration signals through policy-level diversity
- Creating a self-improving exploration curriculum

While optimization stability and memory constraints present challenges, this approach establishes a new dimension for preference-based alignment. Future work should explore hybrid strategies and efficient implementations to realize the full potential of policy-level exploration.

Beyond Static References: Historical Policy Sampling for Exploratory Preference Optimization

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Abstract

Reinforcement Learning from Human Feedback (RLHF) faces fundamental exploration limitations due to reliance on static reference policies for generating comparison responses. While Exploratory Preference Optimization (XPO) introduces theoretical improvements through global optimism, its standard implementation samples comparisons exclusively from the original reference policy. This work develops a framework for adaptive historical policy sampling within XPO, proposing three strategies—GreedyWorst, UniformPast, and DiversityBoost—that dynamically select comparison responses from past policy snapshots. Our approach leverages Low-Rank Adaptation (LoRA) for efficient storage of policy deltas, enabling high-contrast comparisons that could yield more informative preference signals while maintaining constant memory overhead. By transforming the training trajectory into an explorable behavioral repertoire, we address key questions about convergence properties, discovery acceleration, and exploration-computation trade-offs. Although optimization challenges remain, this framework establishes a foundation for more sample-efficient alignment of language models capable of discovering behaviors beyond static human demonstrations.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) has become indispensable for aligning large language models with human preferences Ouyang et al. (2022); Christiano et al. (2023), yet current approaches face fundamental limitations in exploration capability. While offline methods like Direct Preference Optimization (DPO) Rafailov et al. (2024) rely entirely on a static reference policy π_{ref} , even online variants may struggle to discover novel behaviors beyond what the initial policy can produce. Exploratory Preference Optimization (XPO) Xie et al. (2024) represents a significant theoretical advance by incorporating a global optimism term that potentially encourages exploration beyond the reference policy’s support. However, XPO’s standard implementation samples comparison responses $\tilde{\tau}$ exclusively from the original π_{ref} , which could create a bottleneck for discovering diverse, high-quality behaviors.

This work explores how adaptive historical policy sampling might address this limitation within XPO’s flexible framework. While XPO’s formulation permits any sampling strategy for comparison responses (Xie et al., 2024, Section 3), this design space remains largely unexplored in both theory and practice. We investigate specifically how greedy historical sampling - where $\tilde{\pi}^{(t)}$ is dynamically selected from the training trajectory $\{\pi^{(1)}, \dots, \pi^{(t)}\}$ - could enhance the framework. This approach appears to offer compelling advantages: it might maintain only one additional policy snapshot (e.g., the current weakest performer), potentially generate higher-contrast comparisons that yield more informative preference signals, and create an implicit curriculum that continuously challenges the policy to improve.

Our investigation navigates a crucial design space between theoretical exploration breadth and practical efficiency considerations. At one extreme, static reference sampling ($\tilde{\pi}^{(t)} \equiv \pi_{\text{ref}}$) prioritizes stability but may severely limit behavioral diversity. At the opposite end, comprehensive historical sampling (e.g., coverage based selection from all past policies) could maximize coverage but would likely incur prohibitive computational costs. Greedy sampling represents a potentially pragmatic balance, theoretically emphasizing reward maximization through strategic high-value-difference comparisons while maintaining manageable memory overhead. This conceptual framework leads us to consider our fundamental questions: How might adaptive comparison sampling influence XPO’s convergence properties? Could historical sampling accelerate discovery of novel capabilities while preserving training stability? What practical trade-offs exist between exploration diversity, computational requirements, and final performance?

Through this work, we aim to demonstrate how configurable sampling strategies could unlock new dimensions of exploration within the XPO framework, advancing more sample-efficient alignment of language models capable of discovering behaviors beyond static human demonstrations.

2 Related Work

Reinforcement Learning from Human Feedback (RLHF) has established itself as the dominant paradigm for aligning large language models Ouyang et al. (2022), with approaches ranging from proximal policy optimization (PPO) based methods Christiano et al. (2023) to the more recent Direct Preference Optimization (DPO) Rafailov et al. (2024). These techniques share a common limitation: their reliance on passive exploration strategies that sample responses primarily from either a static reference policy or the current policy. This fundamental constraint may hinder their ability to discover novel behaviors beyond the initial policy’s coverage, particularly in the small-regularization regime where theoretical analyses reveal exponential sample complexity challenges Xie et al. (2024).

Recent advances have sought to address these exploration limitations. Exploratory Preference Optimization (XPO) Xie et al. (2024) introduces a global optimism bonus to the DPO objective, creating theoretical pressure to explore beyond the reference policy’s support. Concurrently, online RLHF methods Ouyang et al. (2022) employ iterative feedback collection, while contextual bandit formulations approach preference elicitation as an active learning problem. Despite their innovations, these methods typically sample comparison responses from either a static reference policy or the current policy itself, potentially missing opportunities for richer feedback through diverse behavioral comparisons.

The iterative nature of online RLHF naturally generates a sequence of policy snapshots throughout training, yet most implementations discard these intermediate checkpoints. While iterative DPO Rafailov et al. (2024) shows promise in improving alignment metrics, each iteration typically samples comparison responses from either the original supervised model or the immediately preceding checkpoint. Some approaches introduce specialized explorer networks that generate exploratory responses through controlled perturbations, but these represent lightweight variations rather than true historical snapshots with distinct behavioral characteristics to be used to guide exploration. Our work investigates how XPO’s underutilized flexibility in comparison sampling (Xie et al., 2024, Section 3) could be extended to strategically leverage historical policies as behavioral anchors.

Practical implementation of historical policy reuse would likely require efficient storage solutions. Low-Rank Adaptation (LoRA) Hu et al. (2021) has emerged as a powerful technique for parameter-efficient fine-tuning, compressing policy updates into low-rank matrix factorizations that dramatically reduce storage requirements. While previous applications of LoRA to RLHF have focused on memory-efficient training of single policies, we explore how this technology could potentially maintain a rotating archive of historical policy deltas. This technical approach might enable on-demand reconstruction of previous policies through adapter composition, making policy-level exploration feasible with constant memory overhead. Our framework thus examines how XPO’s theoretically grounded optimism mechanism could be combined with strategic historical sampling, addressing limitations in both algorithmic design and practical implementation.

3 Method

Our methodological approach originates from a fundamental observation: while XPO’s theoretical framework permits sampling comparison responses from any policy, existing implementations default exclusively to the original reference model. This represents a significant unexplored opportunity, particularly given that training naturally generates diverse behavioral snapshots throughout optimization. Constraining comparisons to a static reference may limit exploration breadth, whereas leveraging historical policies could provide richer learning signals. This motivates our framework for adaptive historical policy sampling, where each checkpoint serves as a potential behavioral reference point.

Contrasting current capabilities with historical behaviors creates valuable learning opportunities. Early training policies exhibit high diversity but low competence, while later policies show refinement at the potential cost of exploration breadth. Strategic selection of historical versions for comparison may generate more informative preference pairs. For instance, comparing against less competent predecessors could yield higher-value-difference signals, while sampling from mid-training checkpoints might preserve exploratory behaviors otherwise lost to optimization pressure. This transforms the training trajectory into a dynamic comparison curriculum.

We formalize this through three sampling strategies representing different exploration-exploitation balances:

- **GreedyWorst:** Selects $\tilde{\pi}^{(t)} = \pi \hat{J}_\beta(\pi)$, creating high-contrast comparisons against the weakest historical performer. This approach prioritizes efficient learning signal acquisition while minimizing storage requirements, and forms our primary investigation focus due to its stability advantages and curriculum-like properties.
- **UniformPast:** Samples randomly from all historical policies ($P(\pi^{(i)}) = \frac{1}{|\mathcal{R}|}$), maximizing behavioral diversity but risking low-information comparisons and instability.
- **DiversityBoost:** Employs $P(\pi^{(i)}) \propto \frac{1}{c_i+1}$ to maintain coverage of underrepresented behaviors, potentially preventing or slowing convergence.

The GreedyWorst strategy warrants particular attention as it offers a balanced solution to the exploration-stability tradeoff. By continuously challenging the current policy against its weakest predecessor, it creates a natural curriculum that encourages progress while maintaining training stability. This approach requires storing only one additional policy snapshot—the current worst performer—making it computationally efficient while still providing diverse comparison points as the "weakest" policy evolves throughout training.

Practical implementation necessitates efficient policy storage. We leverage Low-Rank Adaptation (LoRA) to represent policy deltas as compact factorizations $\Delta W^{(i)} = A^{(i)}B^{(i)\top}$ where $A^{(i)} \in \mathbb{R}^{d_{\text{model}} \times r}$, $B^{(i)} \in \mathbb{R}^{d_{\text{layer}} \times r}$ with $r \ll d_{\text{model}}$. Maintaining a fixed-size reservoir $\mathcal{R}$ of these adapters via reservoir sampling enables on-demand reconstruction of historical behaviors. For GreedyWorst, the reservoir simplifies to maintaining only the current policy and the worst historical performer, reducing storage overhead to just two adapter sets.

Algorithm 1 integrates these components, with GreedyWorst as the default sampling strategy. Each iteration samples responses from both the current policy and the weakest historical counterpart. The preference signal feeds into XPO’s objective, which includes an exploration bonus based on historical responses. After policy update, we extract the new behavioral delta and update the reservoir. This elegant extension preserves XPO’s core mechanics while activating its unexplored sampling flexibility.

Algorithm 1 Conceptual XPO with Historical Policy Sampling (GreedyWorst)

Require: T, β, α, N

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1: Initialize  $\pi^{(1)} \leftarrow \pi_{\text{ref}}, \pi_{\text{worst}} \leftarrow \pi_{\text{ref}}, \mathcal{D}_{\text{pref}} \leftarrow \emptyset, \mathcal{D}_{\text{opt}} \leftarrow \emptyset$ 
2: for  $t = 1$  to  $T$  do
3:   Sample prompt  $s_1^{(t)} \sim \rho$ 
4:   Generate response  $\tau^{(t)} \sim \pi^{(t)}(\cdot | s_1^{(t)})$ 
5:   Generate comparison  $\tilde{\tau}^{(t)} \sim \pi_{\text{worst}}(\cdot | s_1^{(t)})$ 
6:   Obtain preference  $y^{(t)} \sim P(\tau^{(t)} \succ \tilde{\tau}^{(t)}) \rightarrow (\tau_+^{(t)}, \tau_-^{(t)})$ 
7:   Update  $\mathcal{D}_{\text{pref}} \leftarrow \mathcal{D}_{\text{pref}} \cup \{(s_1^{(t)}, \tau_+^{(t)}, \tau_-^{(t)})\}$ 
8:   Update  $\mathcal{D}_{\text{opt}} \leftarrow \mathcal{D}_{\text{opt}} \cup \{\tilde{\tau}^{(t)}\}$ 
9:   Optimize:  $\pi^{(t+1)} \leftarrow_{\pi} \mathcal{L}_{\text{XPO}}(\pi; \mathcal{D}_{\text{pref}}, \mathcal{D}_{\text{opt}})$ 
10:  Decompose:  $\Delta W^{(t+1)} \leftarrow \text{LowRankProj}(\theta^{(t+1)} - \theta_{\text{ref}})$ 
11:  Update worst policy:
12:  if  $\hat{J}_{\beta}(\pi^{(t+1)}) < \hat{J}_{\beta}(\pi_{\text{worst}})$  then
13:     $\pi_{\text{worst}} \leftarrow \pi^{(t+1)}$ 
14:  end if
15: end for
16: return  $\hat{\pi} =_{\pi \in \{\pi^{(1)}, \dots, \pi^{(T+1)}\}} \hat{J}_{\beta}(\pi)$ 
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This integration maintains XPO’s theoretical foundations while expanding its practical capabilities. The original convergence properties should extend to historical sampling since we preserve the objective’s structure. For GreedyWorst specifically, the continuous comparison against weaker policies creates a stable curriculum that encourages exploration while avoiding excessive deviation from proven behaviors. The efficiency of maintaining only two policy snapshots (current and worst) makes this approach particularly practical.

We also acknowledge potential emergent dynamics: the optimism term may interact differently with historical samples, possibly necessitating gradient stabilization. However, GreedyWorst’s focus on weak performers likely mitigates this risk by providing consistently high-value-difference comparisons. This strategy represents a paradigm shift from linear progression to curriculum-guided exploration, where historical checkpoints serve as both benchmarks and baselines for improvement and exploration.

4 Experimental Setup

Our experimental framework evaluates historical policy sampling within the XPO paradigm using a two-stage training approach. We first perform Supervised Fine-Tuning (SFT) on the SmolTalk dataset to establish a base policy, then conduct preference optimization using the UltraFeedback binarized dataset. Evaluation is performed on 400 held-out prompts from UltraFeedback to measure instruction alignment quality, with the core metric being win rate against the Qwen2-0.5B-Instruct reference policy, calculated using the Nemotron-70B reward model. For efficiency, we filter all datasets to include only samples up to the 80th percentile of token lengths for computational practicality.

Baseline comparisons include:

- SFT: Base model fine-tuned on SmolTalk
- DPO: Standard Direct Preference Optimization on UltraFeedback binarized

These baselines provide performance benchmarks against which we contextualize the potential of historical sampling strategies:

- *GreedyWorst*: Comparison against weakest historical policy
- *UniformPast*: Random sampling from all historical policies
- *DiversityBoost*: Sampling weighted by inverse selection frequency

Evaluation metrics should examine three dimensions:

- Alignment Quality: Win rate against reference policy
- Behavioral Diversity: Trigram entropy and distinct trigram count
- Training Efficiency: GPU memory footprint

Implementation maintains consistency with these specifications:

Parameter	Value
Base Model	Qwen 2.5 0.5B
SFT Dataset	SmolTalk (filtered: 80th pct length)
Preference Dataset	UltraFeedback binarized (filtered: 80th pct length)
Reservoir Size	up to 5 policies
Optimizer	AdamW ($\text{lr} = 10^{-5}$)
KL Regularization (β)	0.1
XPO Optimism Weight (α)	0.3
Generation	MAX_NEW_TOKENS=512, TEMP=0.6, TOP_P=0.95
Training Duration	1 epoch

All alignment methods train for one epoch on the filtered UltraFeedback binarized dataset. The best checkpoint is selected based on validation win rate, with final metrics reported on the held-out test set.

5 Results

5.1 Quantitative Evaluation

Our baseline experiments establish a performance landscape that highlights opportunities for historical sampling enhancement. The Supervised Fine-Tuned (SFT) model achieved a 34.0% win rate against the Qwen2-0.5B-Instruct reference, indicating moderate but insufficient alignment capability. After Direct Preference Optimization (DPO), performance improved marginally to 35.0%—only a 1.0% absolute gain that underscores the limitations of standard preference optimization. We also observe a 50.5% win rate of DPO against our SFT baseline, demonstrating that while DPO improves upon its initialization, it struggles to surpass the more capable Instruct model.

These results point to two challenges that historical sampling strategies could address. The minimal improvement over Instruct suggests exploration limitations in DPO’s static comparison approach. The narrow margin over SFT indicates lack of behavioral diversity during optimization.

Table 1: Baseline Win Rate Analysis

Comparison	Win Rate	Delta	Interpretation
SFT vs Instruct	34.0%	-	Baseline capability
DPO vs Instruct	35.0%	+1.0%	Marginal improvement
DPO vs SFT	50.5%	+0.5%	Optimization gains over initialization

The modest 1.0% gain against Instruct suggests significant headroom where historical sampling could drive improvement. GreedyWorst’s high-contrast comparisons might amplify learning signals, potentially boosting win rates beyond this threshold. Similarly, the 50.5% win rate over SFT—indicating nearly half of responses show no improvement—suggests DiversityBoost could reduce inconsistencies through broader behavioral coverage. These baseline metrics establish a clear performance and exploration target for historical sampling approaches.

5.2 Qualitative Analysis

A representative example demonstrates DPO’s ability to improve upon SFT while revealing opportunities for further enhancement through historical sampling. Consider the conversion prompt: *“Convert the following sentence with text numbers into digit forms: ‘The library has forty two thousand five hundred sixty seven books in its collection.’”*

SFT Response:

"The sentence can be converted to digit form as follows: '42,567.'"

DPO Response:

"The sentence can be converted into a digit form as follows: 'The library has 42,567 books in its collection.'"

The Nemotron reward scores (-21.375 for SFT vs -17.75 for DPO) confirm DPO’s significant improvement in response completeness and formatting. This 3.625-point gain demonstrates DPO’s effectiveness at learning stylistic preferences. However, both responses focus on surface-level conversion without deeper reasoning, suggesting potential for further enhancement in complex task understanding.

Historical sampling could build on DPO’s foundation to elicit more capable responses. A GreedyWorst comparison against an early policy might contrast with error-rich responses (e.g., "42567 books" without commas), reinforcing correct formatting through higher-value differences. DiversityBoost could encourage more sophisticated pedagogical approaches, potentially yielding responses that demonstrate more reasoning:

Potential Response:

"First, decompose 'forty two thousand' = 42,000 and 'five hundred sixty seven' = 567.

Combining: $42,000 + 567 = 42,567$.

Full conversion: 'The library has 42,567 books.'"

The 50.5% DPO-vs-SFT win rate indicates that nearly half of responses show untapped potential. Historical sampling could potentially increase this through: 1) Strategic policy selection creating more informative comparisons, 2) Diversity preservation maintaining exploratory behaviors, and 3) Behavioral repertoire enabling recovery from plateaus. Our efficient LoRA-based reservoir approach makes this exploration feasible.

This analysis shows that while DPO provides reliable improvements in instruction following, historical sampling could potentially extend these gains to more sophisticated capabilities by transforming the training trajectory into a rich teaching resource. The combination of high-contrast comparisons and behavioral diversity might help bridge the gap to higher reasoning capabilities.

6 Discussion

Our exploration of historical policy sampling within the XPO framework presents a compelling theoretical proposition that would likely encounter significant implementation challenges in practice. The core concept of leveraging policy snapshots as dynamic comparison anchors offers a natural extension to XPO’s flexibility, potentially addressing fundamental exploration limitations in preference-based alignment. However, this approach would likely face several critical constraints that merit careful consideration.

The most significant challenge would likely emerge from optimization instability when combining historical sampling with XPO’s optimism term. We anticipate that processing responses from behaviorally divergent policies could create substantial gradient conflicts, particularly in DiversityBoost sampling where different historical snapshots might generate incompatible optimization signals. This would probably manifest as oscillating loss patterns that hinder convergence, suggesting that practical implementations would require specialized stabilization techniques to handle diverse samples effectively. Alongside optimization challenges, memory constraints would present another substantial barrier. Despite the efficiency gains from LoRA compression, maintaining multiple policy snapshots would introduce considerable GPU memory overhead that could severely limit feasible reservoir sizes, especially for larger models. While GreedyWorst’s minimalist approach of storing just two policies represents the most practical configuration, it would still incur noticeable memory costs compared to standard XPO implementations.

Beyond these technical constraints, broader evaluation challenges would emerge from dependencies on external reward models. Rate-limited access to high-quality evaluators like Nemotron-70B would likely create validation bottlenecks, restricting hyperparameter tuning and progress monitoring. This

dependency highlights the gains to be had from infrastructure development in RLHF research, where advances in optimization techniques benefit from being matched by improvements in accessible evaluation frameworks.

Despite these anticipated challenges, the advantages of historical sampling remain compelling. By decoupling comparison sampling from static references, this approach could alleviate a fundamental constraint in preference-based alignment, while GreedyWorst’s high-contrast methodology might accelerate capability development through targeted challenges. The reservoir mechanism could also preserve valuable exploratory behaviors that standard optimization discards, potentially enabling recovery from training plateaus through access to historical responses. These advantages appear particularly relevant for tasks requiring compositional generalization, where there exist opportunities for capability advancement beyond the stylistic improvements achieved by DPO. The framework further suggests promising applications in continual learning scenarios, where historical snapshots might help mitigate forgetting by preserving behavioral diversity.

Looking forward, successful realization of this approach would require focused efforts in several key areas. Optimization stabilization could resolve conflicts between diverse samples, while novel architectures for memory-efficient policy composition would help transcend current limitations. Also important would be developing more accessible evaluation paradigms through open-source reward models or synthetic feedback systems. The development of this framework underscores the necessity for co-design between algorithmic innovation, hardware capabilities, and evaluation infrastructure. Though implementation hurdles remain, this conceptual exploration establishes a valuable foundation for rethinking how we leverage training trajectories to build more capable and aligned language models.

7 Conclusion

This work establishes historical policy sampling as a promising conceptual advancement for Exploratory Preference Optimization, addressing the exploration bottleneck created by static reference policies in RLHF. By reconceptualizing the training trajectory as a navigable behavioral landscape, we develop a framework that strategically selects comparison anchors from historical snapshots to enable more informative preference learning. The GreedyWorst strategy emerges as particularly compelling, creating a high-contrast curriculum that challenges policies to continuously improve while maintaining minimal storage overhead through efficient LoRA-based reservoirs.

The framework suggests a paradigm shift in how we approach language model alignment—transforming training from linear progression into a multidimensional exploration of behavioral possibilities. This perspective could ultimately enable more sample-efficient discovery of capabilities beyond human demonstrations, balancing acquired skill preservation with novel solution exploration. While implementation challenges around optimization stability and memory efficiency require further investigation, the approach represents a fundamental rethinking of how learning trajectories can be leveraged to build more capable and aligned language models.

Future work should focus on resolving the identified challenges through stabilization techniques and memory-efficient architectures, while exploring promising extensions like transfer learning through behavioral primitives. These developments could unlock the full potential of historical sampling as a new paradigm for preference-based alignment.

References

- Paul Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2023. Deep reinforcement learning from human preferences. arXiv:1706.03741 [stat.ML] <https://arxiv.org/abs/1706.03741>
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685 [cs.CL] <https://arxiv.org/abs/2106.09685>
- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike,

- and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. arXiv:2203.02155 [cs.CL] <https://arxiv.org/abs/2203.02155>
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2024. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. arXiv:2305.18290 [cs.LG] <https://arxiv.org/abs/2305.18290>
- Tengyang Xie, Dylan J. Foster, Akshay Krishnamurthy, Corby Rosset, Ahmed Awadallah, and Alexander Rakhlin. 2024. Exploratory Preference Optimization: Harnessing Implicit Q^* -Approximation for Sample-Efficient RLHF. arXiv:2405.21046 [cs.LG] <https://arxiv.org/abs/2405.21046>